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Q1 Void Function (No Return Value)

5 marks

Write a Python program to define a function that displays "Welcome to Python Programming".

Your Code

```
1 def welcome():  
2     print("Welcome to Python Programming")  
3  
4 welcome()
```

Input

stdin input

Output

```
Welcome to Python Programming
```



Run



Save



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Q2 Void Function (No Return Value)

5 marks

Write a function to print the multiplication table of a given number.

Your Code

```
1 def multiplication_table(n):  
2     for i in range(1,11):  
3         c=n*i  
4         print(c)  
5 multiplication_table(5)
```

Input

5

Output

5
10
15
20
25
30



Run



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Q3 Void Function (No Return Value)

5 marks

Write a function to print all even numbers from 1 to N.

Your Code

```
1 def print_even_numbers(n):  
2     for i in range(1,n+1):  
3         if(i%2==0):  
4             print(i)  
5 print_even_numbers(20)
```

Input

20

Output

2
4
6
8
10
12

▶ Run

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Q4 Fruitful Function (Returns One Value)

5 marks

Write a function that accepts a string and returns the number of vowels in it.

Your Code

```
1 def count_vowels(text):  
2     count=0  
3     for ch in text:  
4         if ch.lower() in "aeiou":  
5             count+=1  
6     return count  
7 result=count_vowels("Hello World")  
8 print("number of vowels:",result)
```

Input

Hello World

Output

number of vowels: 3



Run



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Q5 Fruitful Function (Returns One Value)

5 marks

Write a function that accepts a list of integers and returns the largest element.

Your Code

```
1 def MAX_LIST(a):  
2     c=max(a)  
3     print(c)  
4 a=[10,20,30,40,50]  
5 MAX_LIST(a)
```

Input

stdin input

Output

50

▶ Run

☁ Save



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Q6 Fruitful Function (Returns One Value)

5 marks

Write a function that accepts a number and returns whether it is prime or not.

Your Code

```
1 def is_prime(n):  
2     if n < 2:  
3         return False  
4     for i in range(2, int(n ** 0.5) + 1):  
5         if n % i == 0:  
6             return False  
7         return True  
8  
9 num = int(input())  
10 print("prime" if is_prime(num) else "not prime")
```

Input

7

Output

prime

▶ Run

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SIMATS Online Programming Lab

PYTHON PROGRAMMING



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Your Code

```
1 def marks_total_avg_grade(sub1, sub2, sub3):
2     total = sub1 + sub2 + sub3
3     average = total / 3
4
5     if average >= 90:
6         grade = 'A'
7     elif average >= 80:
8         grade = 'B'
9     elif average >= 70:
10        grade = 'C'
11    elif average >= 60:
12        grade = 'D'
13    else:
14        grade = 'F'
15
16    return total, average, grade
17
18
19 s1 = float(input("Enter marks for Subject 1: "))
20 s2 = float(input("Enter marks for Subject 2: "))
21 s3 = float(input("Enter marks for Subject 3: "))
22
23 total, average, grade = marks_total_avg_grade(s1, s2, s3)
24
25 print("Total:", total)
26 print("Average:", average)
27 print("Grade:", grade)
```

Input

95.5
85.5
75.5

Output

Enter marks for Subject 1: Enter
marks for Subject 2: Enter marks
for Subject 3: Total: 256.5
Average: 85.5
Grade: B



Q8 Function Returning Two or More Values

5 marks

Write a function that accepts the radius of a circle and returns its area and circumference.

Your Code

```
1 import math
2
3 def circle_area_and_circumference(radius):
4     area = math.pi * (radius ** 2)
5     circumference = 2 * math.pi * radius
6     return area, circumference
7
8
9 # Example usage
10 r = float(input("Enter radius: "))
11
12 area, circumference = circle_area_and_circumference(r)
13
14 print("Area:", area)
15 print("Circumference:", circumference)
```

Input

Output

```
Enter radius: Area:
78.53981633974483
Circumference: 31.41592653589793
```



Run



Save



Q9 Function Returning Two or More Values

5 marks

Write a function that accepts a number and returns its square, cube, and square root.

Your Code

```
1 import math
2
3 def square_cube_sqrt(n):
4     square = n ** 2
5     cube = n ** 3
6     sqrt = math.sqrt(n)
7
8     return square, cube, sqrt
9
10
11 num = float(input("Enter a number:\n "))
12
13 s, c, r = square_cube_sqrt(num)
14
15 print("Square:", s)
16 print("Cube:", c)
17 print("Square Root:", r)
```

Input

10

Output

```
Enter a number:
Square: 100.0
Cube: 1000.0
Square Root: 3.1622776601683795
```



Run



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Q10 Keyword Arguments

5 marks

Write a function student(name, age, department) and call it using keyword arguments.

Your Code

```
1 def student(name, age, department):  
2     print("Name:", name)  
3     print("Age:", age)  
4     print("Department:", department)  
5  
6  
7 student(name="Mani Chandan", age=19, department="CSE")
```

Input

Mani Chandan
19
CSE

Output

Name: Mani Chandan
Age: 19
Department: CSE

▶ Run

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